

How to overcome DFM challenges

India-specific strategies for electronics manufacturing include:

- **Building local capabilities.** To boost local capabilities in electronics manufacturing, India must strengthen R&D through public-private partnerships and establish Centres of Excellence and shared labs to drive innovation. Bridging the academia-industry gap through internships, industry-led courses, and mentorship programmes is vital for building a skilled workforce. Additionally, supporting startups via innovation hubs, hardware accelerators, and affordable 'labs on hire' can lower entry barriers and spur grassroots innovation.
- **Policy and ecosystem initiatives.** Offering financial incentives like SPECS can drive domestic component manufacturing and reduce import reliance. Rationalising tariffs on critical imports enhances cost competitiveness for local players. Simplifying compliance procedures and subsidising certification costs can ease the burden on SMEs and startups, encouraging broader participation in high-tech manufacturing through a more supportive regulatory environment.

features such as firmware updates, remote configuration, and diagnostics allow for efficient deployment and maintenance.

Finally, attention to compliance and reliability ensures long-term operability and trust. This includes adhering to relevant safety and EMC standards, such as IEC 61508 and ISO 26262, along with thorough reliability testing, including thermal cycling, vibration, and environmental stress simulations. Choosing components from reputable suppliers with guaranteed availability and documented end-of-life (EOL) timelines strengthens the foundation for a sustainable and certifiable IoT solution.

DFM in practice: Workflow and best practices

A robust DFM workflow is crucial to transforming a concept into a reliable, scalable, and cost-effective electronic product. The end-to-end process begins with clear requirements capture and thorough documentation to align design intent with functional goals. System architecture and block diagrams are then developed to map out major

components and interfaces. Prototyping and proof-of-concept validation follow, serving to de-risk core functionality early in the cycle.

Next, schematic capture and peer reviews ensure design integrity before proceeding to PCB layout, which incorporates DFM checks to catch manufacturability issues early. This is followed by mechanical and thermal validation to confirm fit, heat dissipation, and durability under operating conditions.

The integration of test points, along with detailed production planning, lays the groundwork for streamlined assembly and debugging. Compliance and pre-certification testing ensure that regulatory requirements are addressed upfront, while pilot production enables iterative refinement based on real-world feedback. At every stage, detailed documentation and design traceability support lifecycle management and future upgrades.

Key technical best practices further strengthen the DFM process. These include designing for simplicity and manufacturability, collaborating early with manufacturing and supply chain partners, and

leveraging industry-standard EDA tools equipped with built-in DFM rule checks. Planning for automation and reducing manual intervention not only speeds up production but also enhances consistency. Comprehensive documentation is vital to support traceability, version control, and long-term support. Most importantly, testing must go beyond lab conditions to include real-world scenarios, ensuring that the product performs reliably in its intended environment.

DFM as the foundation for India's electronics future

For India to position itself as a global hub for electronics design and manufacturing, embedding DFM from the outset is not optional—it is essential. DFM is the foundation that enables not just product creation but ecosystem development, fostering trust, resilience, and scalability. Success in this domain hinges on a deep understanding of use cases and real-world constraints, awareness of local supply chain capabilities, and disciplined, structured design thinking. Continuous learning and adaptation are vital as technologies evolve and market demands shift.

By internalising DFM principles and proactively addressing India-specific challenges, the country's electronics sector can move beyond self-reliance to achieve true global competitiveness—delivering world-class products that are both innovative and manufacturable at scale. **EFY**

This article is based on a tech talk session at IEW 2025, Bengaluru, by Mayank Gupta, Founder and CEO, Narnix Edge Pvt Ltd. It was transcribed and curated by Akanksha Sondhi Gaur, Senior Technical Journalist at EFY.

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